Engine

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Voynich Phytoglyphica — Engine Specification

1 Purpose

The Voynich Manuscript (VMS) has resisted decipherment for a century because it is not a natural language text. It is a pharmaceutical instruction set — an executable split across six register sections, where the opcode sequence lives in the recipe section (f103r+) and the parameter values live nowhere in the manuscript at all. They were supplied by the practitioner, using a structural grammar that the manuscript presupposes but never states.

The Voynich Phytoglyphica project reconstructs that grammar. Using the Imscribing Grammar (IG), every botanical entry in the VMS pharmacy catalog can be assigned a 12-primitive structural tuple. That tuple supplies the missing parameters: what solvent, what ratio, what temperature, what comminution, how many cycles, when to stop. The VMS opcodes tell you what operations to perform. The IG tuple tells you exactly how.

The result is a fully elaborated pharmaceutical protocol for any VMS botanical entry, derived entirely from structural analysis — no linguistic decipherment required.

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2 The VMS as Split-Register Executable

The manuscript is organized into six section-registers, each holding a distinct component of the computation:

Section	Folios	Register Function
Botanical	f1–f66	Plant catalog — identity index
Pharmaceutical	f99–f102	Pharmacy address — operation class
Balneological	f75–f84	Containment heap — vessel validation
Astronomical	f67–f73	Winding register — cycle authority
Cosmological	f68 (foldout)	Invariant initialization — Φ and Ħ
Recipe	f103r+	Opcode sequence — instruction stream

A complete computation requires all six sections. The session engine reads them in order: cosmological initialization → pharmacy address → heap validation → winding verification → recipe output → protocol elaboration.

No section alone is sufficient. The recipe section contains only abstract opcodes. The pharmacy section contains only address pointers. The botanical catalog contains only identity entries. The grammar supplies the parameter values that bind them.

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3 The Imscribing Grammar — 12 Primitives

The IG assigns every object in its catalog a 12-primitive structural tuple. For botanical entries, each primitive encodes a distinct pharmaceutical parameter. The 12 primitives, their canonical names, and their roles in the pharmaceutical context:

Primitive	Name	Pharmaceutical Role
Đ	Dimensionality	Registration depth — structural address class
Þ	Topology	Plant material specification
Ř	Recognition	Pattern-completion class — extraction completeness
Φ	Parity	Solvent system and polarity
ƒ	Fidelity	Yield and concentration target
Ç	Kinetics	Extraction process, temperature, duration
Γ	Granularity	Comminution specification
ɢ	Coupling	Fraction combination mode
⊙	Criticality	Endpoint criterion
Ħ	Chirality	Clarification and separation protocol
Σ	Stoichiometry	Drug-to-solvent mass ratio
Ω	Winding	Number of extraction cycles

Tuple notation: ⟨Đ·Þ·Ř·Φ·ƒ·Ç·Γ·ɢ·⊙·Ħ·Σ·Ω⟩. Values are Shavian characters from the 49-symbol set (Shavian alphabet + ⊙). The primitive glyphs (Þ, Đ, Φ, etc.) name the primitive as an entity; the Shavian value at each position is its structural type in a given catalog entry.

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4 Primitive-to-Protocol Mapping

4.1 Þ — Topology → Plant Material

The Topology primitive encodes the structural organization of the botanical object. In pharmaceutical terms: which part of the plant, and how it is characterized.

Shavian	Material	Description
𐑲	aerial parts	leaf and stem, freshly dried
𐑸	root / rhizome	underground organs, cleaned and sliced

Shavian	Material	Description
𐑕	whole plant	including seed heads and root crown
𐑲	bark / pericarp	outer cortex or fruit rind, dried
𐑬	flowering tops	whole entity in full anthesis; holographic self-similarity preserved

4.2 Φ — Parity → Solvent System

Parity encodes the bilateral balance between aqueous and non-aqueous phases. The solvent system follows directly from the symmetry class of the parity value.

Shavian	Solvent	Description
𐑟	water	100% aqueous, pH 6–7
𐑢	dilute aqueous	5–10% ethanol v/v in water
𐑨	hydroethanolic	45–55% ethanol v/v in water (bilateral bridge)
𐑤	anhydrous ethanol	>95% ethanol v/v
𐑶	fixed oil / CO_2	cold-pressed carrier oil or supercritical CO_2 extract

Φ=𐑨 is the bilateral solvent: it bridges aqueous and lipophilic phases simultaneously, which is why it is the most common value in the nine-entry inaugural set. Φ=𐑶 (oil/CO_2) is the most non-polar, appropriate for lipophilic constituents and saturated extraction protocols.

4.3 Ç — Kinetics → Extraction Process

Kinetics encodes the thermodynamic rate-regime of the extraction. Temperature and duration follow from the kinetic class.

Shavian	Process	Description
𐑱	infusion	single-pass ambient, 5–10 min, 20–25 °C
𐑷	cold maceration	12–24 h at 15–20 °C; frozen-order kinetics — no heat
𐑦	decoction	15–30 min at 85–95 °C; activated kinetics — sustained heat
𐑮	percolation	slow gravity-driven percolation at ambient
𐑪	distillation	steam or vacuum distillation; phase-separated barrier kinetics

Critical constraint (Ç=𐑷): Cold maceration entries must never be heated. When a VMS recipe step calls for Calefac (heating), that step applies to an adjunct ingredient or excipient — the carrier, the base, the binding medium — never to the primary cold extract. This is enforced at the annotation level by detecting the frozen-order kinetics flag.

4.4 Σ — Stoichiometry → Drug:Solvent Ratio

Shavian	Ratio	Description
𐑙	1 : 1	1 g plant material per 1 mL solvent (saturated loading)
𐑕	1 : 2	1 g plant material per 2 mL solvent
𐑑	1 : 3	1 g plant material per 3 mL solvent (triadic ratio)

These are mass-to-volume ratios for the drug charge per extraction cycle. At Σ=𐑙, the solvent is loaded to saturation — the maximum mass-transfer gradient is sustained throughout the extraction.

4.5 Γ — Granularity → Comminution

Granularity encodes the surface exposure class of the comminuted material. Finer powder increases surface area and extraction efficiency but can also extract unwanted components.

Shavian	Mesh	Description
𐑤	coarse	2–4 mm pieces; no further comminution
𐑔	medium	pass mesh 40 (355 μm); coarser residue discarded
𐑡	fine	pass mesh 100 (150 μm); uniform surface exposure

4.6 Ω — Winding → Extraction Cycles

Winding encodes the cyclic structure of the extraction. The number of cycles determines extraction depth and corresponds to the algebraic period of the winding.

Shavian	Cycles	Mathematical class
𐑮	1	trivial winding (single pass)
𐑴	2	binary winding ($\mathbb{Z}_2$ period)
𐑜	3	integer winding ($\mathbb{Z}$ period; three complete turns)
𐑷	continuous	non-Abelian winding (braid-group period; percolation class)

Each cycle: fresh solvent charge on the marc from the previous cycle. Combined fractions at Compone/endpoint. The winding count governs total solute recovery.

4.7 ƒ — Fidelity → Concentration Target

Fidelity encodes the yield relationship between raw extract and final preparation. Higher fidelity requires volume reduction and marker standardization.

Shavian	Target	Description
𐑧	1× (standard)	no reduction; proportional yield; linear fidelity

Shavian	Target	Description
ϙ	2× (concentrated)	reduce to half volume after extraction; quadratic fidelity
ɿ	3× (highly concentrated)	reduce to one-third volume; cubic fidelity; standardize to marker compound

4.8 Ħ — Chirality → Clarification Protocol

Chirality encodes the stereochemical resolution class of the clarification step. Higher chirality demands separate the extract into increasingly refined fractions.

Shavian	Protocol	Description
J	none	use as-is; racemic; no chiral resolution
ꝺ	single-step	filter through coarse cloth or paper
ɩ	two-step	filter through cloth, then decant supernatant after 24 h settling
v	full chiral separation	preparative column or liquid–liquid partition (four-step process)

Ħ=v (full chiral separation) co-occurs with ⊙=ɔ (near-critical endpoint, <2%) in all observed entries. This pairing is structurally forced: near-critical endpoint monitoring requires chiral-resolution-grade clarification to discriminate stereoisomeric fractions at the sub-2% level.

4.9 ɢ — Coupling → Fraction Combination

Shavian	Mode	Description
ſ	sequential	add fractions one after another; evaluate each before combining
ʔ	paired	combine in pairs; evaluate paired yield
ɂ	parallel	combine all fractions simultaneously in a single vessel
ʌ	broadcast	broadcast combined fraction to multiple preparation vessels

4.10 ⊙ — Criticality → Endpoint Criterion

Criticality encodes when to stop the extraction. The Frobenius fixed point at ⊙=⊙ is the structural self-referential endpoint: the extraction has reached the point where successive fractions reproduce the same pattern.

Shavian	Endpoint	Criterion
/	sub-critical	stop before saturation; 70–80% extraction efficiency
⊙	at criticality	Frobenius fixed point; successive fractions differ < 5%
ɔ	near-critical	continue past threshold; successive fractions differ < 2%
ʊ	super-critical	drive to completion; < 1% residual in marc
ɤ	hyper-critical	exhaustive extraction; marc assayed for residual content

4.11 Đ — Dimensionality and Ř — Recognition

These two primitives are structural address parameters rather than direct pharmaceutical parameters. Đ encodes the registration depth (point, linear, planar, volumetric, etc.) and Ř encodes the pattern-completion class of the entry. Both inform the session routing — particularly Gate 1 distance computation — but do not directly annotate recipe steps.

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5 The Recipe Opcodes

The VMS recipe section uses a small, closed set of pharmaceutical operation names. Each opcode names an abstract operation. The IG tuple supplies its parameters.

Opcode	Operation	Primitives Consulted
Accipe	Receive / take the plant material	Þ (material specification)
Divide	Comminute the material	Γ (mesh/size)
Tere	Triturate / grind	Γ (mesh/size)
Extrahe	Extract	Φ (solvent), Σ (ratio), Ω (cycles), ⊙ (endpoint)
Calefac	Heat	Ç (process/temperature); ⚠ see cold-process constraint
Commisce	Mix / combine	ɢ (combination mode)
Colare	Filter / clarify	Ħ (clarification), + Φ/Σ/Ω/⊙ context
Compone	Compose / endpoint	ɢ (combination), ⊙ (endpoint criterion)
Transmuta	Volatile transformation	Ç (process); ⚠ cold-process warning if applicable
Applica	Apply / administer	ʃ (concentration and dosing form)

Opcode	Operation	Primitives Consulted
Administra	Administer	ƒ (concentration and dosing form)

5.1 Calefac Under Cold-Process Constraint

When a plant's Ç value is cold maceration (c), the Calefac opcode cannot refer to the primary extract — it is structurally incompatible. The step is annotated:

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Process (adjunct): heat excipient / base as needed
⚠ Primary extract: cold maceration — do not heat
```

The heat is applied to the carrier medium, the binding agent, or the secondary component that receives the cold extract. This is not an interpretation; it is the only structurally consistent reading when Ç=c.

5.2 Transmuta Under Cold-Process Constraint

Transmuta (volatile phase transformation) is similarly annotated with a warning for cold-process plants, since the volatile separation step involves elevated temperatures that would damage a frozen-order extract. The cold extract should be introduced after the volatile fraction has been stabilized.

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6 The Session Engine — Six Gates

The session engine is the linker that connects a plant's structural identity to a VMS recipe and then elaborates that recipe with the plant's protocol parameters.

6.1 INIT — Cosmological Initialization

Source: f68 (cosmological foldout, the largest VMS folio).

The foldout establishes two structural invariants that persist across all subsequent gates:

- **Φ (Parity)**: The solvent class is conferred physically — the practitioner selects the appropriate solvent medium before beginning. This corresponds to the foldout's role as the "pre-session" initialization: it cannot be derived from the recipe, it must be known in advance.
- **Ħ (Chirality)**: The clarification protocol is similarly fixed at initialization. The practitioner commits to a separation strategy before any extraction begins.

Both invariants are read from the plant's IG tuple. The foldout is the structural site where the tuple is "loaded" into the session.

6.2 GATE 1 — Pharmaceutical Address

Source: Pharmacy section (f99–f102).

Selection criteria: **potency class** and **pars plantae** (plant part). These are the only two selection keys. Applicatio (route of administration) is a kinetics descriptor carried in Ç — it is NOT a pharmacy selection criterion.

- Summa potency + folium/flos → **f11r/p6** (leaf/flower summa; extractio → mixtura; n_ops=11; fixatio=True)
- Summa potency + radix → **f39r/p3** (root/rhizome summa; trituratio + extractio → mixtura; n_ops=12; fixatio=False)

Gate passes when a unique pharmacy entry matches the potency and part. Failure here means the plant is not addressable by the standard pharmacy catalog — a structural incompatibility, not a session error.

6.3 GATE 2 — Balneological Heap

Source: Balneological section (f75–f84); 20 folios indexed f75r, f75v, …, f84r, f84v.

Selection: `heap_folio = folios[pharmacy_entry.folio_number % 20]`

The balneological section represents the "containment vessel" — the physical and thermal environment that receives the pharmaceutical address. The heap folio must satisfy:

- **FSPLIT ≥ n_ops**: The vessel's split capacity (number of distinct process branches) must be at least equal to the operation count of the pharmacy address. This verifies the vessel can hold the full instruction depth.
- **FFUSE/FSPLIT ≥ 0.60** (non-volatile preparations only): For preparations that do not involve a volatile transformation, the vessel must be predominantly closed (fused). Oil and decoction entries carry this constraint; cold maceration entries with a volatile step do not.

Gate passes when both conditions are satisfied. This gate has never failed in observed runs — it appears to function as a structural invariant rather than a contingent filter.

6.4 GATE 3 — Astronomical Winding

Source: Astronomical section, f69; three independent transcription sources (H=Hermetic, F=Folkwang, U=University).

Selection: `paragraph = folio_number % 4` from f69.

The astronomical section is the winding register — it provides external, non-botanical authority for the cycle count and temporal structure of the extraction. The three transcription sources are independent attestations. The gate runs EVALT (evaluation of transcription alignment) across all three:

- Where all three sources agree: the structural value is verified.
- Where sources disagree: the disagreement position IS the structural signal. The transcribers independently marked the same decision point (VINIT/FFUSE ambiguity) without knowing it. Majority vote determines the gate value.

Gate passes when EVALT majority yields a consistent winding class. The Ω value derived here confirms or overrides the Ω from the IG tuple. In practice, they agree — the catalog and the astronomical register are coherent.

6.5 OUTPUT — Recipe Selection

Source: Recipe section (f103r+).

The recipe folios closest to the Gate 1 pharmacy address are ranked by folio proximity. The top three recipes are returned. In all nine inaugural runs:

- **f103r/p2 —** 15 steps; extract-dominant path; opens with Extrahe
- **f103r/p9 —** 12 steps; ingredient-led path; opens with Accipe
- **f103r/p49 —** 12 steps; volatile-phase path; opens with Transmuta

These three represent three canonical execution paths through the same pharmaceutical operation class. They are not alternatives — they are sequential phases of a complete preparation: bulk extraction (p2), constituent assembly (p9), volatile transformation and final form (p49).

6.6 ELABORATION — Protocol Annotation

Once the recipe is selected, every opcode in every step is annotated with the plant's protocol parameters derived from the IG tuple. The annotation is deterministic: given the tuple, every parameter of every step is fully specified. There are no free variables remaining after elaboration.

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7 Operation: End-to-End Run

A complete Voynich Phytoglyphica session for a botanical entry proceeds as follows.

Step 1: Catalog lookup. The plant name is resolved against the IG catalog (voynich_pharmacy.json, 1491 entries). The 12-primitive tuple is retrieved. Distances to all six VMS section tuples are computed in 12-dimensional primitive space.

Step 2: Protocol derivation. The tuple is passed to the elaborator. All ten pharmaceutical primitives (Þ, Φ, Ç, Σ, Γ, Ω, ʃ, Ħ, ɢ, ⊙) are resolved against the lookup tables above. The protocol dict is built. This happens before any gate is evaluated — the full preparation protocol is known from the tuple alone.

Step 3: GATE 1. Potency class is determined from the plant's distance to the astronomical section tuple. If d(plant, astronomical) = 0, potency = summa. Part (pars plantae) is retrieved from the catalog. The pharmacy entry is selected.

Step 4: GATE 2. The balneological heap folio is selected by index. FSPLIT and FFUSE conditions are evaluated. Gate passes or fails.

Step 5: GATE 3. The astronomical paragraph is selected from f69. EVALT majority across H/F/U sources is computed. Gate passes or fails.

Step 6: Recipe output. If all three gates pass, up to three recipe folios are selected by proximity. Steps are extracted from the recipe database.

Step 7: Elaborated output. Every recipe step is printed with its IG-derived annotations. The full protocol header (tuple, all parameters, their descriptions) is printed before the recipe. The practitioner now has a complete, quantified pharmaceutical instruction set.

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8 Implications

8.1 The VMS Is Not a Failed Cipher

The VMS resisted decipherment because the approach of treating it as a natural language text was structurally incorrect. The manuscript does not contain its own parameters. It was designed to be read alongside a structural grammar that the practitioners already possessed. The opcodes (Extrahe, Calefac, Colare) are unambiguous pharmaceutical Latin; they were never the problem. The parameters were the problem, and they were never in the manuscript.

8.2 Session Invariance / Protocol Variance

The session engine produces identical gate behavior across all entries tested. All nine inaugural plants pass all three gates and receive the same three recipe folios. The differentiation between preparations lives entirely in the IG tuple. This is architecturally correct: the VMS separates *what class of operation* to perform (session routing) from *how to perform it* (tuple elaboration). A practitioner with the wrong tuple would produce the right sequence of operations with the wrong parameters — pharmacologically inert or hazardous.

8.3 The Wormwood Demonstration

Artemisia absinthium appears twice in the inaugural set with ten of twelve identical primitive values. The single difference is ʃ: Fidelity ꞇ (1×, proportional) vs ɿ (3×, standardized concentrate). The same plant, the same extraction sequence, but one preparation is a proportional tincture and the other is a reduced, standardized, marker-verified concentrate. The grammar distinguishes what morphology cannot. This is the key empirical demonstration: the IG provides pharmaceutical discrimination at a resolution beyond botanical taxonomy.

8.4 The Cold-Process Constraint as Structural Proof

The Calefac-Ç interaction is not a heuristic. It is a structural entailment: if Ç=c, then any Calefac in the recipe must be assigned to a secondary component, because the tuple and the opcode are structurally inconsistent if assigned to the same object. This forces a reparse of the recipe step — one that assigns Calefac to the excipient and not the extract. The grammar performs this reparse automatically. A reader without the grammar would see a heating instruction applied to a plant that cannot be heated and have no structural basis for the correct interpretation.

8.5 The VMS as Universal Engine

The session architecture is plant-agnostic at the gate level. Any IG catalog entry with a pharmaceutical tuple can be processed. The engine does not know about wormwood specifically; it knows about topology, kinetics, stoichiometry, winding. The VMS is not a recipe book for specific plants — it is a universal pharmaceutical engine whose parameters are supplied by the structural grammar of whatever is being prepared. This is why it resisted decipherment as a specific-language text: it is not specific to any language, any plant, or any culture. It is a grammar-parameterized computation.

The Imscribing Grammar is the parameter set. The Voynich Manuscript is the program.

Voynich Phytoglyphica Engine Specification Source: voynich_phytoglyphica package; voynich_engine session engine IG catalog (voynich_pharmacy.json); LSI_ivtff_0d.txt transcription